# The cortical column as a tuned receiver: a network mechanism for temporal-interference stimulation

Abstract and storyline


## Abstract

Temporal-interference (TI) stimulation applies two high-frequency currents whose superposition contains no field component at their low difference frequency. Neural populations can generate that component by nonlinear mixing and amplify it selectively by recurrent resonance. In a Jansen–Rit column, the firing-rate sigmoid mixes the carriers through its curvature. In an exact quadratic integrate-and-fire mean field, the corresponding static curve emerges when the membrane variables relax faster than their input while the synapses remain dynamic; at faster inputs, mixing depends on the population dynamics and cannot be read from the steady-state curve. Fast cellular access and slow circuit resonance can therefore coexist in one population model. In both descriptions the leading weak-input beat is proportional to the product of the effective carrier amplitudes, and the membrane's attenuation of the carrier enters that product once, through those amplitudes. Recurrent dynamics near a Hopf bifurcation amplify the beat as $1/\gamma$, where $\gamma$ is the damping of the critical mode. Connectivity controls this gain at fixed generated drive, and beat-locked rate modulation grows faster than the mean-rate change, in the mean field and in a spiking network of 8000 neurons per population. Because the beat follows the carrier product rather than the envelope excursion, an envelope-only dose map rests on a physiological assumption. TI and direct low-frequency stimulation share the downstream amplifier, so resonance alone confers no TI efficiency advantage. The biological access pathway and the absolute response at human field strengths remain unresolved.


The abstract is the TN abstract of v0.14.4, proposed on 3 October 2026 and awaiting K's approval; the JNE build keeps a structured abstract with the same claims. The map below records the v0.14.3 organization, which v0.14.4 leaves unchanged; the decisions on the last page are the structural questions the map raised. Nothing is cut or reordered before approval.

## Problem and theory

First argument block: Sections 1–2

A two-tone field carries no power at its beat frequency, so a neural response there requires a nonlinear element. The question is how much of that response belongs to the population circuit rather than to the cell. The theory builds the receiver in three stages, access, mixing, and amplification, and shows that the static and dynamic detectors are one object seen at different carrier speeds.

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| **§1** Introduction | The two-tone field (Eq. 1) has lines only at the applied tones; a linear system cannot create the beat. Cellular mixing exists (MRG and Hodgkin–Huxley axon models; Luff's subthreshold mixer at 5 kHz, TTX-sensitive). Network dependence exists (timing-not-rate in primates; pyramidal responses lost under synaptic block; network simulations). The paper separates carrier access, beat generation, and recurrent amplification, and the population reduction does not identify the anatomical access route. | States the problem and the distinction the paper turns on: cellular mixing versus the network response it recruits. Announces the ladder (JR column, QIF mean field, spiking network, laminar column) once. The transfer curve must be defined before its curvature can mix, so §2.1 follows. | Fig. 1 (receiver concept). Grossman 2017; Mirzakhalili 2020; Luff 2024 (mixer); Vieira 2024; Caldas-Martinez 2024; CFC and HAM companion papers. |
| **§2.1** Population transfer and the slow-synapse limit | JR synaptic filter and sigmoid (Eqs. 2–3); NMM1 with the direct field term $\Lambda[\hat{E}]$ (Eq. 3). The MPR mean field (Eq. 4) and its static curve $F(Q)$ (Eq. 5), inflection at $Q = \Delta/\sqrt{3}$. The slow-synapse reduction (Eq. 6) and its two conditions: $\tau_{\mathrm{rel}} \ll \tau_s$ makes the transfer static for synaptic input; $\omega_c \tau_{\mathrm{rel}} \ll 1$ is the stronger condition for a carrier. The QIF reduction does not contain sodium-channel kinetics. Cellular and circuit resonance are distinct. | Defines what a transfer curve is and when it is static, so that curvature can be a mixer. Introduces the time-scale distinction that the Discussion and Conclusion reuse. The field must reach the curve, which §2.2 supplies. | Rosetta Stone (Ruffini 2025); Montbrió 2015; Clusella 2023; de Palma Aristides 2026 (WP0242, unpublished); Hutcheon and Yarom 2000. |
| **§2.2** From applied field to a squared-envelope beat | Quasistatic field; polarization $s(t)$ through parallel entry points with access filters (Eqs. 7–8), one compartment response counted once. Taylor expansion (Eqs. 9–10) with the truncation condition $\lvert s\rvert \ll 1/\rho \approx 1.8$ mV. The square of the envelope (Eq. 11): $A^2$ is sinusoidal, $A$ is a rectified cosine. The AM surrogate (Eq. 12), where it is used and where it is not. Three predictions: $\varepsilon^2$ scaling, $\Sigma''$ law, carrier independence at fixed polarization. | Derives the beat and its amplitude: the first result. Fixes the waveform discipline (two tones for every waveform-sensitive claim). The derivation assumes a static curve, so §2.3 asks what a dynamic population does. | Figs. 3 and 6 (two-tone runs); Fig. 6d (surrogate check). HAM paper (Volterra kernel); Galan and Gadea 2023. |
| **§2.3** The exact mean field connects static and dynamic mixing | $C(Q,\omega)$ (Eq. 13) with limits $F''/4$ and $F'/(2\omega^2)$ (Eq. 14), sign change at $\omega_z$ above the inflection. Fixed-excursion coefficient $K(\omega)$ (Eq. 15): the $1/\omega^2$ is the membrane, not a detector filter. The PING $v_E^2$ source matches the second-order response to 1.1%, 0.1%, 0.03% at 300, 1000, 2000 Hz. Carrier and beat take different routes to the rate (Table 1). | Shows that the static and dynamic detectors are one object at different carrier speeds, with membrane access counted once; settles the detector-penalty question of September 2026. Whatever the detector, its envelope law has observable signatures, which §2.4 derives. | Fig. 7; Table 1. Code: `static_vs_dynamic_check.py`, `mpr_twotone_lines_check.py`. |

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| **§2.4** Transduction class and its signatures | Cycle average (Eq. 16) is even in $A$; square law is generic for a smooth transfer; rectifiers follow $A$; a step detector follows neither. Harmonic ladder 1/5, 3/35, 1/21 and $r_{\mathrm{det}}$ (Eq. 17). The network suppresses the second harmonic ($\mathcal{Q} = 10$: 20% reads 0.7%), so the test runs off resonance at $\Delta f \simeq f_0/4$. Excursion map (Eq. 18): product law, 2× versus 40× example; not an account of focality. Spatial response (Eq. 19) and signed vector coupling. | Converts the detector class into measurements (amplitude slope, harmonic ratio, dose map) and separates spatial from dynamical selectivity. Generation gives a weak drive; §2.5 explains what makes it large. | App. D (bound and step detector); Fig. 6b (slope 1.99); Table 3 (predictions). |
| **§2.5** Amplification: synaptic low-pass and network resonance | Synaptic transfer $H_s$ (16 Hz corner; 0.7 at 10 Hz, 2.5% at 100 Hz). One-mode reduction, $\mu_\pm = -\gamma \pm i\omega_0$, $\mathcal{Q} = \omega_0/2\gamma$, gain $\propto 1/\gamma$ (Eq. 20); entrainment above the Hopf. Receiver equation (Eq. 21) with $D$ and $G$ and their normalization for JR and QIF. The cubic normal form has no quadratic mixer. | Supplies the second factor of the receiver and the claim that a large susceptibility is not evidence of detection. The carrier must still reach the mixer, which §2.6 quantifies. | Fig. 5; Figs. 4 and 9; App. C. |
| **§2.6** Carrier access depends on the effective membrane response | $H_m$ (Eq. 22); $\lvert H_m\rvert^2 \approx 10^{-4}$ at 1 kHz for 16 ms, 0.39 and 0.14 at 1 and 2 kHz for 0.2 ms, labeled as assumed. Table 2 (response stages). AC space constant (Eq. 23), one tenth at 1 kHz. A gate filter needs its own model; the QIF coefficient is not a calibrated sodium response. Carrier sweeps constrain access plus mixing; beat sweeps probe the amplifier. | Separates carrier independence at fixed polarization from the applied-field roll-off and makes fast access a labeled hypothesis. The detector coefficient also depends on state, which §2.7 takes up. | Figs. S5, S6, S9; Deitcher 2017; Eyal 2016; Discussion paragraph on Cohen, Luff, Semenov. |
| **§2.7** Operating-point dependence | PEIX (Eq. 24) tracks $\Sigma'$ and peaks where $\Sigma''$ vanishes; the complementarity belongs to the memoryless detector. The dynamic detector at a fast carrier follows $F'$ with no sign rule. State shifts (neuromodulation, pathology) scale, and in JR can flip, the response. | Turns the detector coefficient into a brain-state prediction and marks which part is model-specific. The theory is complete; Methods specify the simulations that test it. | Fig. 2b; Fig. 6a; CFC paper (PEIX). |

JR, Jansen–Rit; MRG, McIntyre–Richardson–Grill; QIF, quadratic integrate-and-fire; MPR, Montbrió–Pazó–Roxin; NMM1/NMM2, sigmoid and exact mean-field neural mass models; PING, pyramidal–interneuron network gamma; AM, amplitude modulation; PEIX, Population Excitation–Inhibition Index; CFC, cross-frequency coupling; HAM, hierarchical amplitude modulation.

## Simulations

Second argument block: Sections 3–4

The Results climb the ladder. The JR column verifies the three predictions of §2.2; the exact mean field reproduces generation and amplification with a derived nonlinearity; the spiking network confirms the timing signature at the microscale. Methods stand apart because the lock-in read-out and the pinned-operating-point protocol recur across models.

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| **§3.1** Jansen–Rit: integration, lock-in, controls | Field enters only the pyramidal sigmoid; the read-out $v = y_1 - y_2$ is not fed the field. RK4 steps with $< 1\%$ convergence; settling and windows. Hopf at $p \approx 315$, $f_0 \approx 11.1$ Hz; SNIC at $p \approx 114$. Lock-in (Eq. 25) with DC subtraction to $10^{-8}$. Open-loop detector versus closed-loop column; square law, linearization, finite-difference checks. | Makes every JR number reproducible and defines the lock-in that every later section uses. | Fig. 2; Grimbert and Faugeras 2006; `run_all.py`. |
| **§3.2–3.4** LaNMM, NMM2 PING, QIF network | LaNMM: field as a signed polarization, not a clipped rate (a clip is a half-wave rectifier); alpha-power read-out; signed maps. PING (Eq. 27): weights $(1, 1, 1, 2)$, $\bar{\eta}_H = 1.110$, $f_0 = 54$ Hz, field as current $I_E = PE$ so the membrane computes access with $\tau_m = \tau_E$; two criticality knobs. QIF: $N = 8000$ per population, Lorentzian quantiles, beat 55 Hz (forced) and 42 Hz (gated, 11 Hz detuning); modulation depth and pairwise phase consistency; the 45% threshold sensitivity of phase concentration. | Fixes the models whose results follow and states which comparisons are perturbative. Decision for K: §3.2 serves only the App. E results and could travel with them. | Eqs. 26–27; Fig. S2 (LaNMM setup); Rosetta Stone PING parameters; Vinck 2010. |
| **§3.5–3.6** Coupling sweeps, tACS control, vocabulary | Pinning: solve $p(C)$ for fixed $v^*$, read $\gamma, \omega_0$ from the Jacobian; tACS control at $\varepsilon = 0.1$ mV. Three terms: forced response, entrainment (dominance screen $\lvert f_{\text{out}} - \Delta f \rvert < 0.3$ Hz), gating. | The pinning protocol isolates amplification from detection; the vocabulary keeps "entrainment" from covering gating. | App. F (metrics); Pikovsky 2001; Canolty 2010; Tort 2010. |
| **§4.1–4.2** The sigmoid demodulates; the beat drives a resonance | An alpha line from two tones with no alpha power in the input. Lorentzian response growing $0.23 \to 0.43 \to 0.70$ mV toward the Hopf ($1/\gamma$); above the Hopf, Arnold-tongue entrainment read by output frequency. | The existence result: the beat is synthesized and amplified. | Figs. 3, 4; Fig. S4. |
| **§4.3–4.4** Carrier independence, kHz roll-off, curvature law | Flat response over 25–300 Hz at fixed polarization; $1/f_c^2$ roll-off for 16 ms, retained access for 0.2 ms; two tones at 2 kHz with $2.2 \times 10^{-13}$ input leakage. Curvature law to 0.9%; slope 1.99; linearization collapses the response 634×; two-tone/AM ratio 1.0000 at 0.02 mV and 0.988 at 1 mV. | Verifies the three predictions of §2.2 and licenses the surrogate for the dense sweeps only. | Figs. 5, 6; Figs. S6, S9; Esmaeilpour 2020. |

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| **§4.5** Exact mean-field confirmation | PING generates and amplifies the beat through $v_E^2$; diagonal ridge with operating point; the detector sits on the fast side of its crossover, with a source operating-point independent to about 1%. | The receiver is not an artifact of the sigmoid. | Figs. 7, 8; Fig. S8; Clusella 2023. |
| **§4.6** Coupling sets the gain at fixed detection | Pinned $J$-curve: $\sim 30\times$ growth, $1/\gamma$ then saturation; the same gain–$\gamma$ relation via $p$; stable-focus side only. NMM2 $J$-curve without pinning. | The mesoscale claim: amplification depends on distance to criticality, not on which parameter delivers it. | Fig. 9; Fig. S10; Clusella 2023. |
| **§4.7–4.8** Timing over rate; QIF confirmation | Modulation grows 13.5× (26 to 346 mHz) against a mean shift of 5.8× (−1.3 to −7.7 mHz); ratio 19 far from the Hopf, 45–66 near it. QIF: bunching below the Hopf; gating with sidebands at 95 and 11 Hz above it; modulation depth 0.003 → 0.29/0.23 with rate +8–10%; weak per-neuron locking. | The timing-not-rate signature at mesoscale and microscale, qualitatively matching Vieira. | Fig. 10; Figs. S11, S12; Vieira 2024. |
| **§4.9** Falsifiable predictions | Four predictions with tests: access versus mixing by carrier sweep; state and carrier dependence; transduction class by amplitude slope and harmonic ratio off resonance; frequency selectivity. Table 3 against existing evidence; synaptic block tests network involvement without excluding a cellular detector. | Converts the mechanism into falsifiable statements and bridges to the Discussion. | Table 3; Vieira 2024; Caldas-Martinez 2024; Opancar 2025. |

LaNMM, laminar neural mass model; tACS, transcranial alternating-current stimulation; SNIC, saddle-node on an invariant circle; RK4, fourth-order Runge–Kutta.

## Interpretation and supporting material

Third block: Sections 5–6 and Appendices A–G

The Discussion changes level from model results to physiology and states each scope limit at the claim it limits. The appendices hold the notation, the model equations, the two bounds, the laminar extension, and the metrics.

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| **§5** Discussion | Generation versus amplification does not compete with cellular mixing. The time-scale distinction and what the QIF reduction omits. Fast-access origins (nodes, AIS, terminals; RC independent of area). Cohen's periaxonal space ($\sim$ 1 ms fitted decay), Luff's 1 ms RC, Semenov's 0.7 $\mu$s charging: a 10–100$\times$ shorter time is a hypothesis. Dose map: min versus product and the DC term. Intensity changes the mechanism. Shared amplifier with tACS; temporal, not spatial, criticality. Couplings 0.12–0.5 mV per V/m; $\eta_{\mathrm{det}} \simeq \lvert\Sigma''\rvert\varepsilon/4\Sigma'$; $\mathcal{Q} \sim 10$. Vieira's 0.34 (0.04–0.65) against 0.8% and why that is not a rejection. The 0.348 supremum and its scope. Limitations: Plovie 2025, noise, columns, cascaded synapses cannot solve access, mixed human results. | Reinterprets each result in physiological terms and places every caveat where it bites. Retain all paragraphs; each adds a level (mechanism, time scales, anatomy, dosimetry, intensity, efficiency, scale, data, bound, limits). | Fig. S3 (tACS); Cohen 2020; Luff 2024 (PWM-TI); Semenov 2025; Vieira 2024; App. D. |
| **§6** Conclusion | Receiver: mixing generates, resonance amplifies, and a large response does not imply efficient detection. Static and dynamic descriptions connected by time scales; each filter once. Product law, shared amplifier, open access pathway. | Restates the storyline in three paragraphs without new claims. | — |
| **App. A–B** Notation and units; model equations | Table 4 (symbols, with the reused-glyph head note); JR equations (Eq. 28) and Table 5. | Recovery aids; definitions remain local. | — |
| **App. C–D** Cubic normal form; efficiency bound | Stuart–Landau has no second-order source (Eq. 31); numerical control at the floor. $\eta_{\mathrm{det}} \leq \rho\varepsilon/4$ (Eq. 32); family supremum 0.3479 at $\delta/2\varepsilon = 0.523$ (Eq. 34), a step detector with harmonic ratio 0.151; the microvolt width is an idealization, and the maximum is numerical, not certified. | C supports §2.5 (susceptibility is not detection); D supports §2.4 and the Discussion's efficiency paragraphs. | `threshold_bound_check.py`. |
| **App. E** Laminar column at low carriers | Tongues at 25–80 Hz carriers up to 3 mV, outside the perturbative regime; 95%/97% of the $P_1$-driven grids below baseline; double resonance under superficial drive (7% off resonance) against broadband response under deep drive (25%); three LaNMM predictions, the carrier-matching one untested. | Shows a band-limited intermediary makes the carrier choice matter; does not carry the kHz argument. | Fig. S1, S2, S7; Gordon 2019. |

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| **App. F–G** Vocabulary metrics; supplementary figures | Dominance screen versus capture test; folded-rate metric; Arnold-tongue protocol. Figures S2–S12 support §4.2–4.8 and the Discussion. | Metrics behind the three terms; the figure bank for the JNE supplement. | Figs. S2–S12. |

**Decisions for K.** (1) Theory holds seven subsections and about 4,500 words before any simulation; §2.4 (transduction class) and §2.6 (carrier access) serve the predictions and the Discussion more than the Results. Keep the current order, or move those two after the Results as interpretation sections. (2) Methods §3.2 (LaNMM) describes an experiment whose results are in App. E; in the JNE article the method appears without its result. Keep, or move §3.2 into App. E. (3) Figures S3 (tACS shares the amplifier), S10 (NMM2 $J$-curve), and S11 (timing against rate) support main-text claims in §4.6–4.7 and the Discussion from the supplement; the main text has ten figures. Promote none, one, or all three. (4) The abstract above replaces the v0.14.3 TN abstract; approve or revert. None of these is executed in v0.14.4.